DATA ANALYST PORTFOLIO

Superstore Sales Analysis: SQL-Based Revenue, Customer, and Regional Performance Insights

Transforming Raw Survey Data into Actionable Insights Through SQL Business Analysis, Excel Visualizations, and Storytelling

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Tools & Technologies:
Excel | SQLite/DB Browser| SQL | GitHub

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1. **Summary**

This project analyses the Sample Superstore sales dataset using SQLite for querying and Excel for visualization to uncover revenue performance, regional contribution, customer concentration, and monthly sales patterns.

The analysis demonstrates an end-to-end business analytics workflow: data import, SQL aggregation, result extraction, chart creation, and insight communication.

The final output includes four polished charts: Top 10 Product Categories by Revenue, Monthly Sales Trend, Sales by Region, and Top 10 Customers by Sales.

This project demonstrates how SQL-based sales analysis can support product prioritization, regional strategy, customer retention, and trend monitoring

2. **Business / Analytical Question**

This project was designed to answer four key business questions:

1. Which product categories generate the most revenue?
2. How does sales performance change over time?
3. Which regions contribute the most revenue?
4. Which customers drive the highest sales value?

3. **Dataset**

The project uses the Sample Superstore CSV dataset, imported into SQLite as an orders table. The dataset includes order-level fields such as Order Date, Customer Name, Region, Category/Sub-Category, Sales, Quantity, Discount, and Profit, making it suitable for revenue, time-series, and customer-level analysis.

4. **Tools Used**

- SQLite/DB Browser for SQLite- for data import and SQL querying
- SQL- for aggregation, grouping, sorting, and business question answering
- Excel- for charting and output formatting
- GitHub- for project publishing and portfolio presentation.

5. **Process (What I Did)**

    1. Downloaded and extracted the Sample Superstore CSV dataset.
    2. Created a local SQLite database and imported the CSV as an orders table rather than manually defining columns. This avoided table-definition errors and enabled immediate SQL analysis.
    3. Wrote SQL queries to calculate:
        - total sales
        - top product categories by revenue
        - monthly sales trend
        - sales by region
        - top customers by sales
    4. Exported/copied the query results into Excel as compact result tables rather than exporting the entire raw table, which was an issue earlier in the workflow.
    5. Built four charts in Excel:
        - Top 10 Product Categories by Revenue
        - Monthly Sales Trend
        - Sales by Region
        - Top 10 Customers by Sales
    6. Interpreted the outputs for business meaning and prepared them for GitHub portfolio presentation.
    7. *The SQL analysis used SUM, GROUP BY, ORDER BY, LIMIT, and date transformation logic to answer core business questions.*

6. **Key Visualizations**

1. Top 10 Product Categories by Revenue

The best-performing categories are led by Phones and Chairs, followed by Storage, Tables, and Binders. Phones generated $330,007.05, while Chairs generated $328,449.103.

2. Monthly Sales Trend

The line chart shows monthly sales from 2014-01 onward, with uneven performance over time and recurring peaks across the analysis period.

3. Sales by Region

Regional revenue is led by:

1. West: $725,457.825
2. East: $678,781.24
3. Central: $501,239.891
4. South: $391,721.905

4. Top 10 Customers by Sales

The customer chart shows a concentration of value among a relatively small number of buyers. The top customers include:

- Sean Miller: $25,043.1
- Tamara Chand: $19,052.2
- Raymond Buch: $15,117.3
  with the rest of the top 10 all above roughly $12,000 in sales value.

7. Key Insights

1. The project's total sales calculation returned $**2,297,200.8603**, giving the business a clear top-level revenue view before segmentation.
2. Product revenue is concentrated: **Phones** and **Chairs** are almost tied at the top, while the broader top 10 is dominated by office, furnishing, and technology-related sub-categories.
3. The **West** region is the strongest geographic market, outperforming the **South** by a wide margin.
4. Customer value appears concentrated: a relatively small group of customers contributes a large amount of revenue, which suggests an opportunity for key-account strategies and retention analysis.
5. Monthly sales are not flat; the corrected monthly trend shows variation across time, supporting the idea of seasonality, campaign timing, or uneven order behaviour across periods.

8. Business Implications and Recommendations (Why This Project Matters)

This project matters because it reflects the sort of practical questions commercial teams ask every day:

- Which categories deserve more focus?
- Which customers are most valuable?
- Which regions should receive more strategic investment?
- When does performance rise or weaken?

A manager could use this analysis to support:

- Inventory prioritization
- Category promotion
- Regional targeting
- Customer retention or loyalty segmentation
- Monthly forecasting and operational planning

The project proves that you can take a raw dataset and turn it into structured, actionable business insight.

9. Challenges and How to Solve Them

Challenge 1: Incorrect export from SQLite

Initially, exporting through File → Export → Tables as CSV exported the entire orders table rather than the smaller query result table, which created the wrong structure in Excel. The solution was to copy the actual result set with headers from the SQL results panel and paste only the compact outputs into Excel.

Challenge 2: Date parsing for monthly sales

The monthly trend query originally returned NULL for the month because the Order Date values were stored as text in a non-ISO format, and the initial SQLite parsing attempt failed. I corrected this and ultimately produced a valid month sequence in the final Excel output from 2014-01 onward.

Challenge 3: Safe file/database setup

Saving the database in Program Files failed due to Windows permissions, so it had to be saved in a user-controlled project folder instead. That is a practical workflow lesson for future local SQL work.

10. Portfolio-Ready "What I would Improve Next."

To make the project even stronger in a production setting, I would:

- Add profitability analysis (not just revenue), using the Profit field.
- Build a dashboard version of this project in Cognos or Excel.
- Add customer segmentation (e.g. high-value vs low-value customers)
- Extend the time-series analysis to include moving averages or quarter-over-quarter trends.
- Reproduce the same project in Python/Jupyter Notebook to strengthen cross-tool portfolio depth.

11. Files and Links

Chart 1: Monthly Sales Trend

Chart 2: Sales by Region

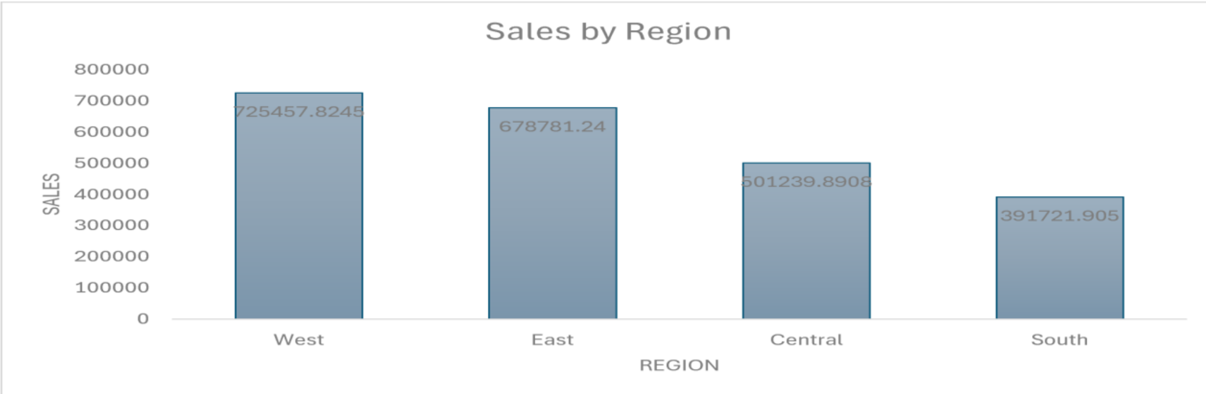


Chart 3: Top 10 Product Categories by Revenue

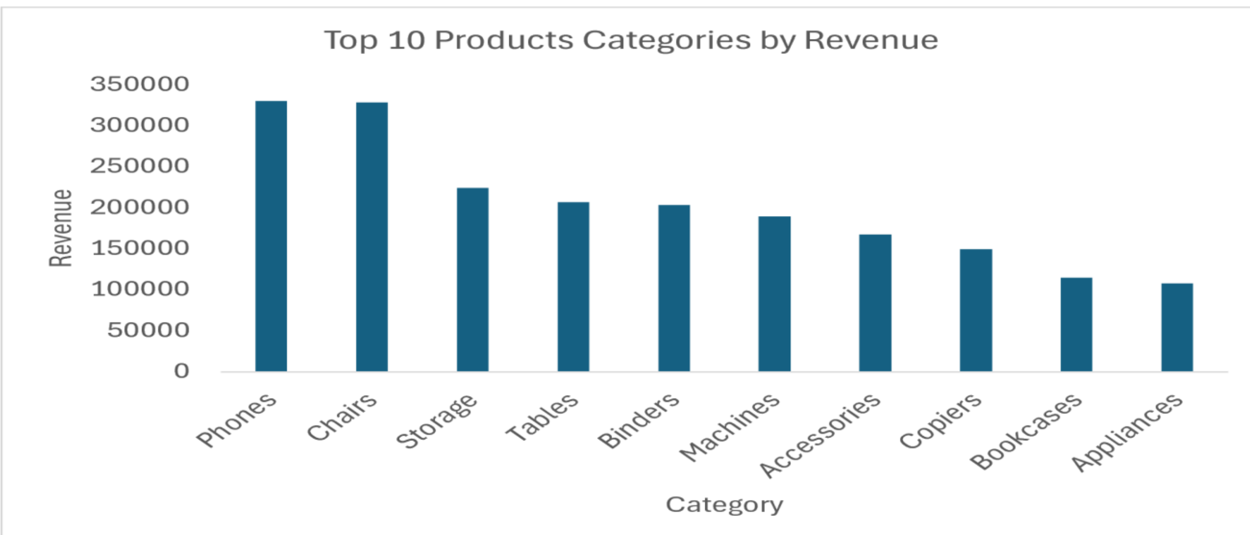
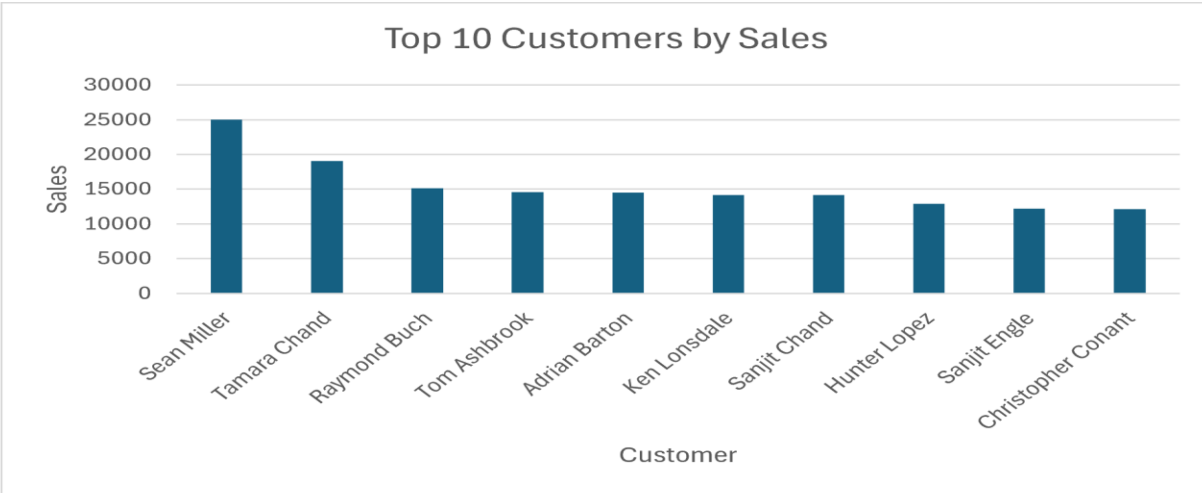


Chart 4: Top 10 Customers by Sales


Top 10 Customers by Sales
Sales
30000
25000
20000
15000
10000
5000
0
Sean Miller
Tamara Chand
Raymond Buch
Tom Ashbrook
Adrian Barton
Ken Lonsdale
Sanjit Chand
Hunter Lopez
Sanjit Engle
Christopher Conant
Customer